

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru | github.com/zac-09 | Kampala, Uganda (UTC+3) — Remote

PROFESSIONAL SUMMARY

Senior Backend Engineer building **public APIs** and **platform systems** on Node.js, TypeScript, MongoDB, and Kubernetes. Led the end-to-end, zero-downtime migration of 20+ MTailor production apps from Parse/MongoDB to Firebase/GCP, designing a real-time MongoDB↔Firestore Pub/Sub **integration platform** that supported parallel production traffic during cutover. Comfortable shaping data models, evolving service boundaries toward monorepos, and turning ad-hoc backends into reliable, observable systems. Fluent in async-remote work since 2020 across US, Singapore, and Uganda timezones, and in agentic AI-assisted workflows (Claude Code, Playwright). Open-source: github.com/zac-09.

CORE COMPETENCIES

Public APIs & Extensibility Platforms

Node.js & TypeScript

MongoDB

Microservices & Monorepo

AWS & Kubernetes

Distributed Systems

Zero-Downtime Migrations

API Design & Data Modeling

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote

- Led the **end-to-end migration of 20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Architected and implemented a real-time two-way **MongoDB↔Firestore sync pipeline** on Node.js + Google Pub/Sub — an internal integration platform delivering eventual consistency across services during cutover.
- Migrated file storage from Amazon S3 to GCS via Python; consolidated services off AWS and saved the company **\$5,000/month** in recurring infrastructure cost.
- Built a 3D customer visualisation feature (video overlay + ffmpeg), increasing buyer conversion; shipped Express Shipping generating \$40 additional revenue per order.

Stack: Node.js, TypeScript, MongoDB, Firebase, GCP Pub/Sub, AWS S3/EC2/EBS, Python, React

MTailor

Mar 2022 – Jul 2022

Backend Contractor · US Remote

- Hired via Upwork to execute a complex backend migration from Parse/MongoDB to Firebase Firestore and GCP; built migration scripts, processing logic, and parallel-write strategies.
- Wrote engineering documentation for Firebase SDK adoption (Node.js + Python) and trained the Ops team on the new Firebase dashboard and backup procedures.

Stack: Node.js, MongoDB, Firebase, GCP, AWS, JavaScript

CodeBits

Jan 2020 – Jul 2021

Software Engineer · Tech Lead · Remote

- Architected a **microservices backend** for FIDA Uganda's case management platform using Apache Kafka, Docker, and Kubernetes — a multi-user, eventually-consistent system used across NGO chapters.
- Designed and shipped a **gRPC-based service bus** exposing legal-aid backend functionality to third-party USSD providers — an early extensibility surface with explicit schema versioning and backward-compat guarantees.
- Led a team of 4 developers; established code review culture, paired pull requests, and written architecture notes.
- Delivered a cross-platform mobile app for LASPNET (React Native + Expo, iOS + Android) with Firebase push notifications.

Stack: Node.js, Apache Kafka, NATS Streaming, gRPC, Docker, Kubernetes, React Native, Firebase

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Built ad-hoc batch jobs querying **2M+ Firestore records** to keep customer stock prices current from Morningstar APIs — cursor-paginated, idempotent, predictable in cost.
- Extended backend APIs using Firebase Cloud Functions and Express.js hosted on Heroku; added responsive UI pages with React + Tailwind.

Stack: Node.js, Firebase, React, Tailwind

Mind2matter.co

Jan 2022 – Jul 2022

Full-Stack Engineer · US Remote

- Built DeFi backends (Web3 + Node.js) and delivered React UIs from Figma under tight client timelines; reported to the CTO.

Stack: Node.js, Web3, React, Firebase

PROJECTS

Career-Ops — Agentic Job-Search Platform [Open Source](#)

AI-assisted pipeline built on Claude Code + Playwright: scans 45+ portals, evaluates offers, generates tailored CVs, tracks applications. Demonstrates fluency with modern AI tooling integrated into a real engineering workflow.

Tech: Node.js, Claude Agent SDK, Playwright, YAML, Puppeteer PDF generation

MongoDB ↔ Firestore Real-Time Sync Pipeline [Production](#)

Two-way live sync pipeline (Node.js + Google Pub/Sub) supporting parallel production traffic during the MTailor Parse→Firestore migration. Designed for idempotency, conflict resolution, and rollback at the service boundary.

Tech: Node.js, MongoDB, Firestore, Google Pub/Sub, GCP

gRPC Service Bus for USSD Legal-Aid Providers [Production](#)

Versioned gRPC interface exposing legal-aid backend functionality to multiple independent USSD providers across Uganda — an early extensibility-platform pattern with schema-first contracts.

Tech: Node.js, gRPC, Protocol Buffers, Apache Kafka, Kubernetes

EDUCATION

B.Sc. Computer Engineering · Makerere University

Aug 2018 – Sep 2022

Kampala, Uganda

SKILLS

Proficient: Node.js, TypeScript, MongoDB, Firebase, React, GCP, Python, Linux, SQL, Git

Working knowledge: Go, AWS, Rust, GraphQL, React Native, Kubernetes, Apache Kafka, gRPC

AI-assisted dev: Claude Code, Playwright agents, agentic workflows **Languages:** English (fluent), Kiswahili